CM25D008

PRACTICAL NO :-08

AIM :- A system has RAM which can store four pages simultaneously to satisfy
page requests. Write a Program to calculate percentage page hit and page fault if
the system uses Optimal page replacement technique for the page references
entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3)

MINGW64:/c/Users/sejal sarode/OneDrive/Desktop

```
  GNU nano 8.7
#include <stdio.h>

int main() {
    int frames = 3;
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int n = 12;

    int frame[3];
    int i, j, k, pos, farthest;
    int pageFaults = 0, pageHits = 0;

    // Initialize frames
    for(i = 0; i < frames; i++)
        frame[i] = -1;

    for(i = 0; i < n; i++) {
        int found = 0;

        // Check HIT
        for(j = 0; j < frames; j++) {
            if(frame[j] == pages[i]) {
                found = 1;
                pageHits++;
                break;
            }
        }

        // PAGE FAULT
        if(!found) {
            pos = -1;
            farthest = i + 1;

            for(j = 0; j < frames; j++) {
                int k;
                for(k = i + 1; k < n; k++) {
                    if(frame[j] == pages[k])
                        break;
                }

                if(k == n) { // Not used again
                    pos = j;
                    break;
                }
```

MINGW64:/c/Users/sejal sarode/OneDrive/Desktop

GNU nano 8.7

```c
        // PAGE FAULT
        if(!found) {
            pos = -1;
            farthest = i + 1;

            for(j = 0; j < frames; j++) {
                int k;
                for(k = i + 1; k < n; k++) {
                    if(frame[j] == pages[k])
                        break;
                }

                if(k == n) { // Not used again
                    pos = j;
                    break;
                }

                if(k > farthest) {
                    farthest = k;
                    pos = j;
                }
            }

            if(pos == -1)
                pos = 0;

            frame[pos] = pages[i];
            pageFaults++;
        }
    }

    printf("Total Page Hits = %d\n", pageHits);
    printf("Total Page Faults = %d\n", pageFaults);

    float hitRatio = (float)pageHits / n * 100;
    float faultRatio = (float)pageFaults / n * 100;

    printf("Hit Percentage = %.2f%%\n", hitRatio);
    printf("Fault Percentage = %.2f%%\n", faultRatio);

    return 0;
}
```

```
sejal sarode@INBOOK_X2_SLIM MINGW64 ~/OneDrive/Desktop
$ nano optimal.c

sejal sarode@INBOOK_X2_SLIM MINGW64 ~/OneDrive/Desktop
$ gcc optimal.c -o optimal

sejal sarode@INBOOK_X2_SLIM MINGW64 ~/OneDrive/Desktop
$ ./optimal
Total Page Hits = 5
Total Page Faults = 7
Hit Percentage = 41.67%
Fault Percentage = 58.33%
```